

Lecture 16: Strong Law of Large Numbers, Chernoff Bound

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- 1 Beyond convergence in probability::: **Theorem: Strong Law of Large Numbers** If $X_1, X_2, \dots$ iid mean μ : $\bar{X}_n \rightarrow \mu$ almost surely, i.e. $P(\lim_n \bar{X}_n = \mu) = 1$ – stronger than convergence in probability.::: **Theorem: Chernoff Bound** For $t > 0$, Markov applied to e^{tX} gives $P(X \geq a) \leq e^{-ta} M_X(t)$; optimizing over t :

$$P(X \geq a) \leq \min_{t>0} e^{-ta} M_X(t),$$

typically decaying *exponentially* – far tighter than Chebyshev.::: **Chebyshev vs. Chernoff, head to head** 1

—title: “Lecture 16: Strong Law of Large Numbers, Chernoff Bound”author: “Jamie Haddock”format: revealjs: output-file: Lecture16_slides slide-number: true preview-links: auto logo: figs/hmc.png css: input/slides.css incremental: true smaller: true code-fold: true embed-resources: true html: code-fold: true embed-resources: true pdf: documentclass: article toc: true number-sections: true geometry: - top=1in - left=1in - bottom=1in - right=1informat-links: falsejupyter: python3filters: - input/remove-pause.luaexecute: echo: true eval: true—

- 1 Beyond convergence in probability::: **Theorem: Strong Law of Large Numbers** If $X_1, X_2, \dots$ iid mean μ : $\bar{X}_n \rightarrow \mu$ almost surely, i.e. $P(\lim_n \bar{X}_n = \mu) = 1$ – stronger than convergence in probability.::: **Theorem: Chernoff Bound** For $t > 0$, Markov applied to e^{tX} gives $P(X \geq a) \leq e^{-ta} M_X(t)$; optimizing over t :

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```
n = 100S = rng.choice([-1, 1], size=(200000, n)).sum(axis=1)a_values = np.arange(5, 45, 5)true_probs =
```

—# Case Study: Optimal Sampling for Randomized Matrix MultiplicationBack in Week 5 we sampled $p_i \propto \|a_i\| \|b_i\|$ without justifying it. One can show (Cauchy-Schwarz, minimizing variance over the simplex) this is exactly the **variance-minimizing** choice – an optimization-meets-probability moment – and it also gives sharper concentration than uniform sampling.

```
n_, m, p_dim = 6, 60, 5
A = rng.normal(size=(n_, m))
B = rng.normal(size=(m, p_dim))
true_AB = A @ B
def sample_AB(n_iter):
```

We now leave static random variables behind and start building random *processes*.
 Reading & Homework: §5.5, *supplementary notes (Bertsekas & Tsitsiklis, Introduction to Probability)*.
Due today: Homework 7 (Gradescope).